

16 methods and comparators over 56 evaluations; blue clears that column's floor, orange falls below.

method (grouped by role)	cell-context												donor				unseen perturbation						
	T1 · Kang									T5c · OP3			T2	T3 · CRISPR					T4	T5u			
ctrl-pred	-.02	.00	.02	.00	.01	-.02	.01	.00	-.03	.01	-.01	-.01	.00	-.01	.00	-.02	.00	.00	.02	.00			
cell-mean floor	.67	.62	.62	.71	.53	.46	.53	.58	.07	.02	.00	.01	.26	.52	.59	.46	.49	.45	.66	.17			
donor-shift	.83	.79	.72	.82	.60	.82	.88	.78	.28	.19	.26	.32	.02	.52	.59	.46	.49	.45	.66	.17			
linear-PCA floor	.83	.78	.72	.82	.60	.82	.88	.78	.30	.18	.26	.33	.04	.39	.45	.08	.30	.29	.28	.15			
scGen	.77	.78	.77	.73	.54	.84	.82	.75	.13	.16	.24	.19	.15										
CPA/chemCPA	.36	.34	.53	.50	.27	.41	.26	.41	.11	-.02	.10	.04	.21							.10			
biolord														.37	.46	.32	.53	.32	.61	.13			
GEARS														.19	.34	.01	.35	.17	.44				
AttnPert														.21	.32	.13	.38	.15	.50				
scGPT	.67	.64	.62	.69	.54	.40	.54	.58	.38	.28	.35	.38	.25	.39	.33	.00	.29	-.17	.57				
scFound.	.75	.73	.68	.70	.52	.69	.69	.65	.28	.28	.38	.38	.12	.23	.25	.10	.08	-.01	.30				
STATE	.59	.63	.68	.70	.41	.76	.79	.58	.08	.00	.03	.05	.14	.10	.18	.05	.33	.05	.23	.12			
PertAdapt														.17	.40	.23	.27	.13	.52				
CellOT	.81	.82	.76	.80	.56	.91	.90	.74	.15	.15	.26	.23	.37										
scPRAM	.80	.73	.69	.81	.54	.90	.86	.68	.19	.12	.22	.20	.16										
PerturbNet	.79	.79	.78	.82	.55	.94	.88	.75	.12	.10	.22	.15	.25	.32	.43	.29	.50	.32	.60	.16			
PRnet									.13	.04	.08	.13								.13			
CellFlow	.83	.82	.79	.83	.56	.93	.88	.77	.10	.07	.09	.15	.24	.34	.17	.30	.44	.38	.49	.17			
FP-ridge									.42	.31	.39	.44								.16			
lin-shift-KO														.51	.60	.43	.49	.45	.61				

cell fill = margin over that column's binding floor (Pearson-Δ)

-.43

.00

+.13

blue = above the floor; white = at the floor; orange = below it

clears both floor members and survives multiplicity correction

point-estimate clearance, without statistical support

adapted interface

diagnostic comparator

not evaluated

row accent:

baseline

conditioned

diagnostic comparators

93 of 106 donors
over the cell-mean floor
mean gap +0.107 [+0.086, +0.128]
paired Wilcoxon $p = 1.6 \times 10^{-13}$

mean +0.107

cell-mean floor

(Pearson- Δ)

donor (Soskic CD4 leave-one-donor-out), sorted by gap